

Build Instructions

„xOTA Antenna“ 100W

DG1JAN

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Safety Notice

These assembly instructions are intended for licensed radio amateurs who, in accordance with their qualifications, have been instructed on the dangers of electrical current, can assess these risks, and can ensure safe operation.

The antenna is designed solely for temporary use in dry weather conditions (portable radio operation). Maximum power is 100W SSB and CW, or 50W for FT8 or similar operating modes. High RF voltages can occur in antennas (even at low transmitting powers). Appropriate safety clearances and protection against accidental contact must be maintained.

Disclaimer

The creator of these instructions assumes no liability for damage to equipment or personal injury! Further claims are excluded.

No liability is accepted for direct, indirect, or consequential damages resulting from the replication of this project. This applies equally to damages resulting from improper or unintended use.

BOM

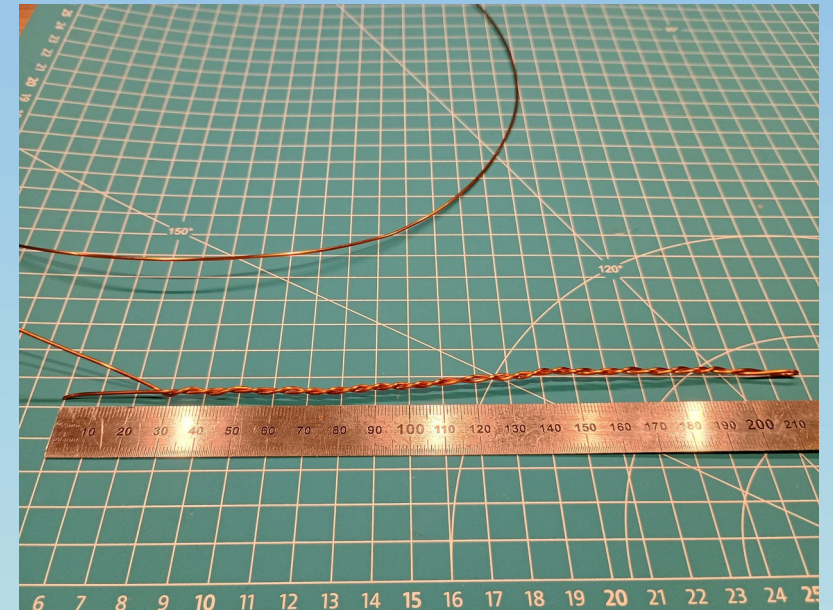
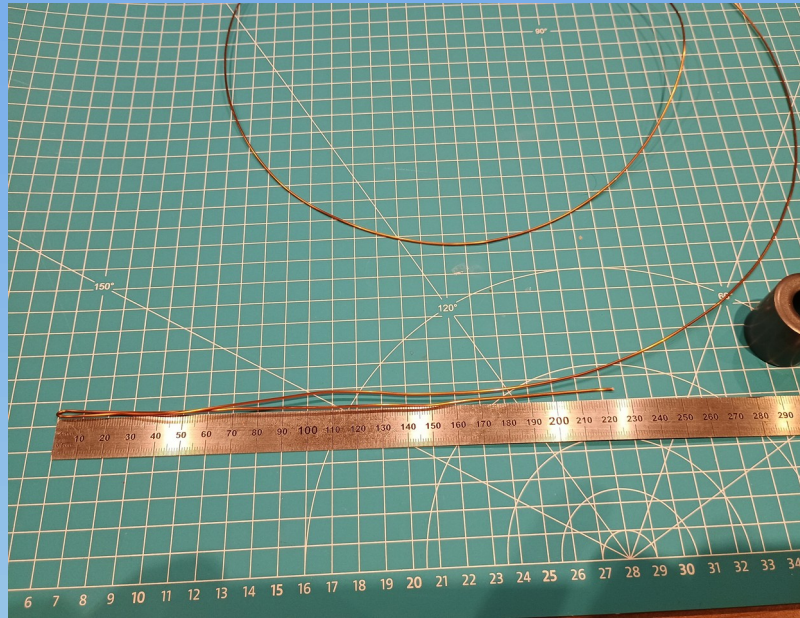
- 1x PCB xOTA-Antenna 100W
- 1x Toroid FairRite 2643101902
- 1x Ceramic Capacitor 100pf 1kv
- 1x BNC Socket Amphenol B6252H7 (or similar)
- 1x 3D Printed wedge
- 1x Cabletie 2,5mm/150mm
- 2x 4mm „Bana“ Sockets
- 1,6m enamelled copper wire 1mm/AWG 18
- 1x Heatshrink PVC 85mm

optional:

- 1x 3D printed Coil winder

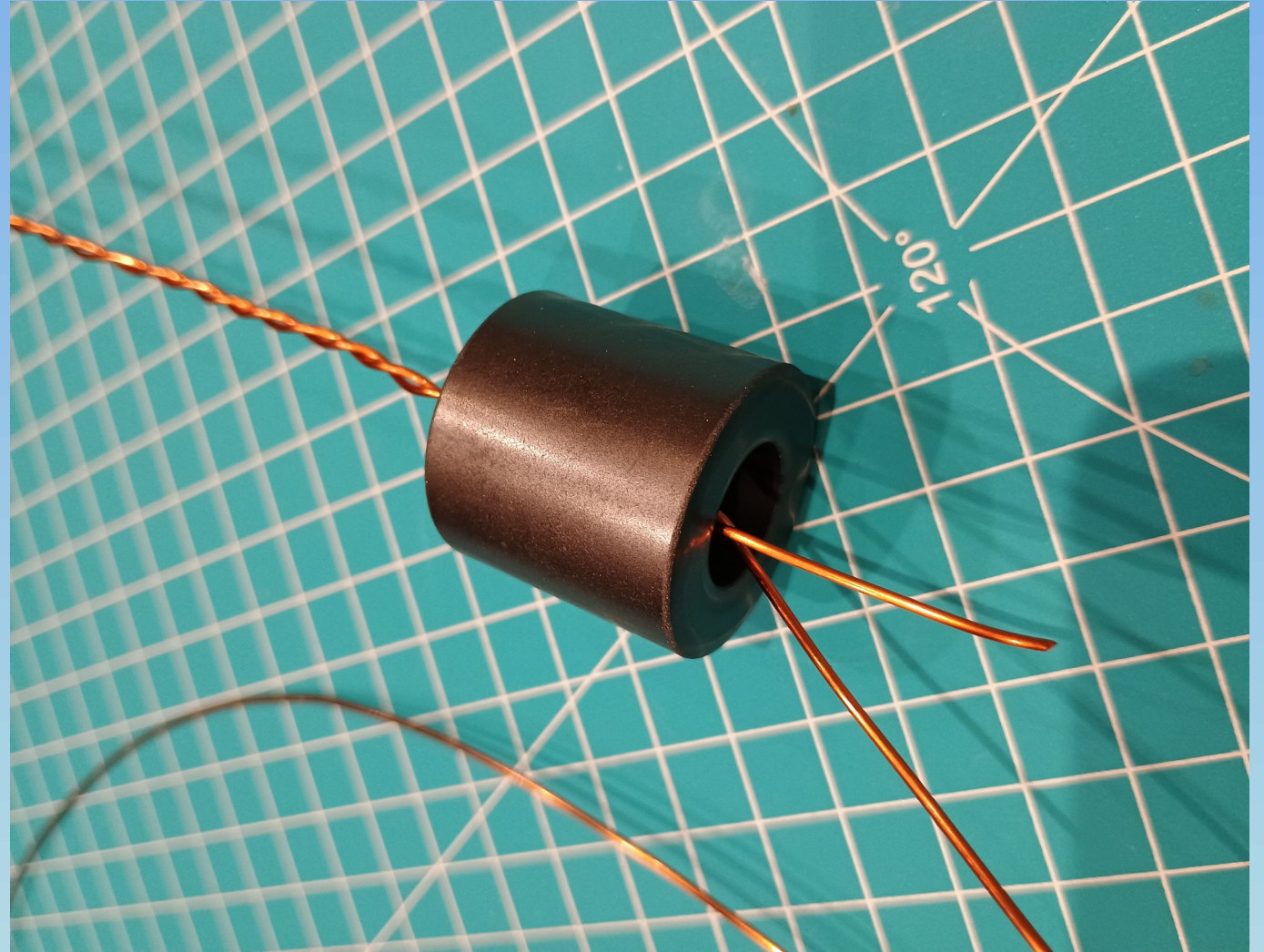
1) Built the Trafo

- Fold enamelled copper wire at one end ~22cm back
- Twist wire and let 3-4cm untwisted on the end



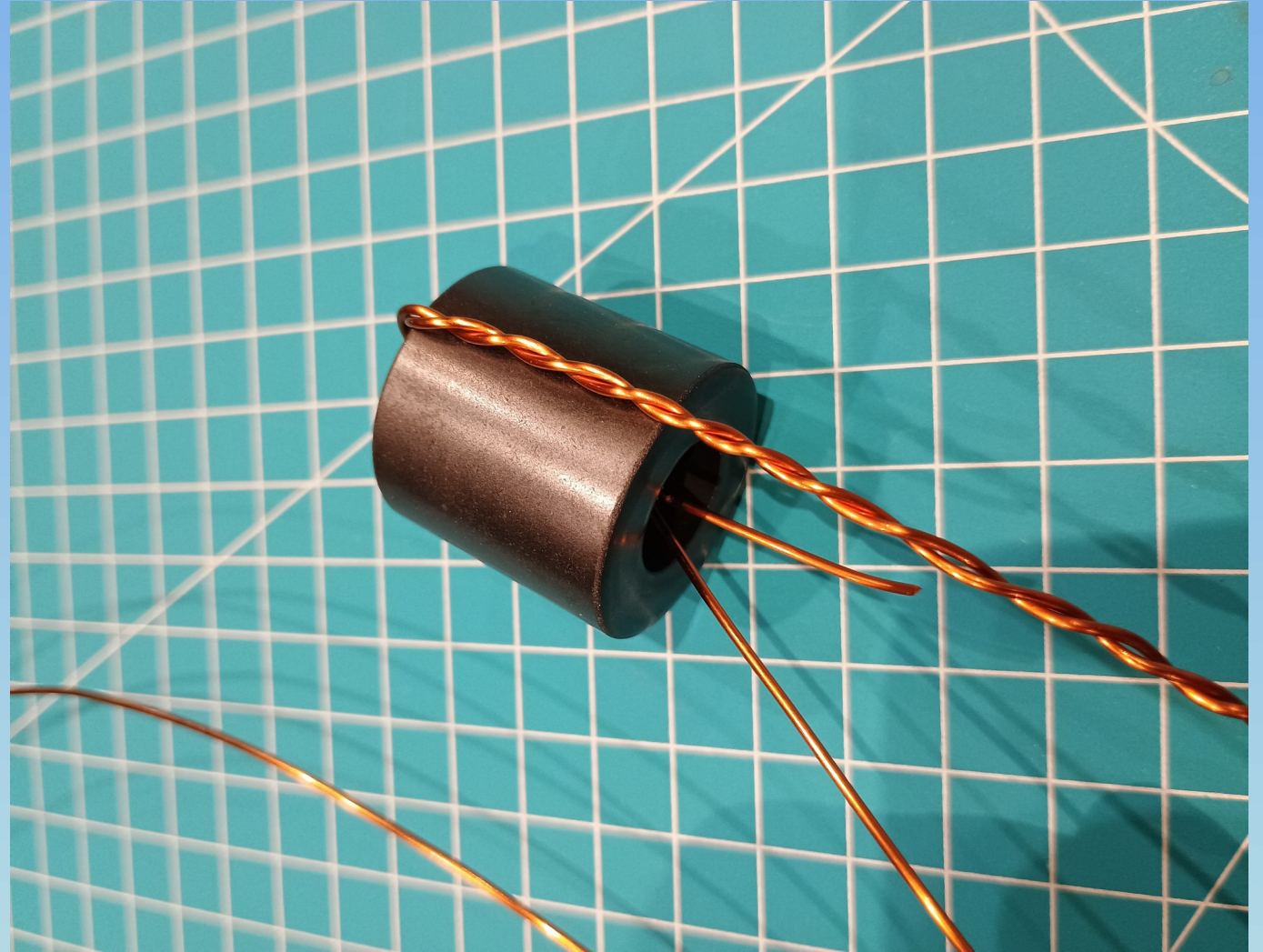
1) Built the Trafo

- Put the wire inside the torroid as shown in the picture



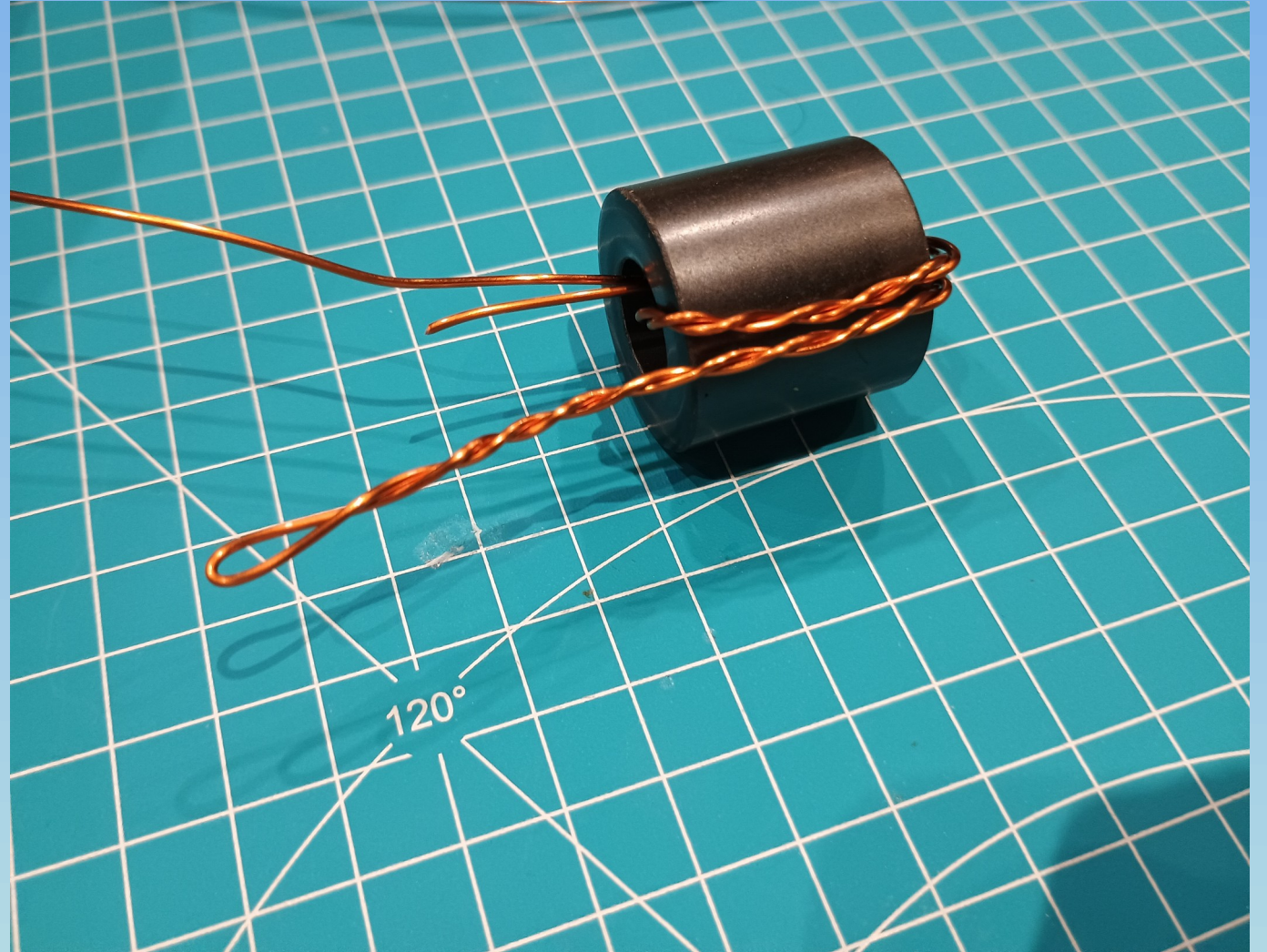
1) Built the Trafo

- Wind the twisted part of the wire to the right side of the toroid



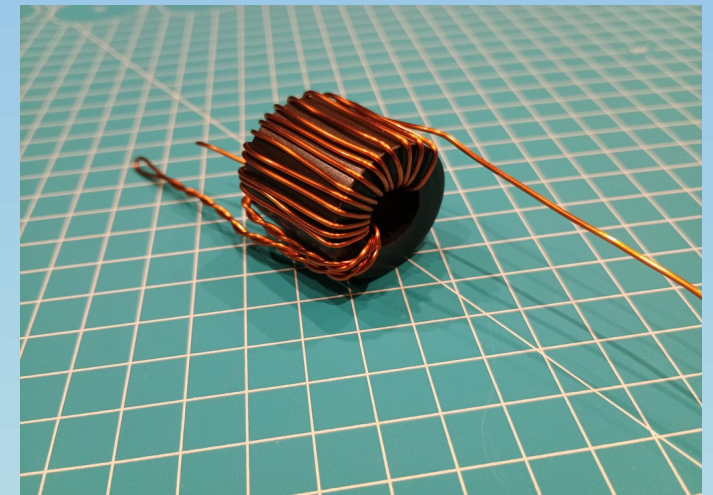
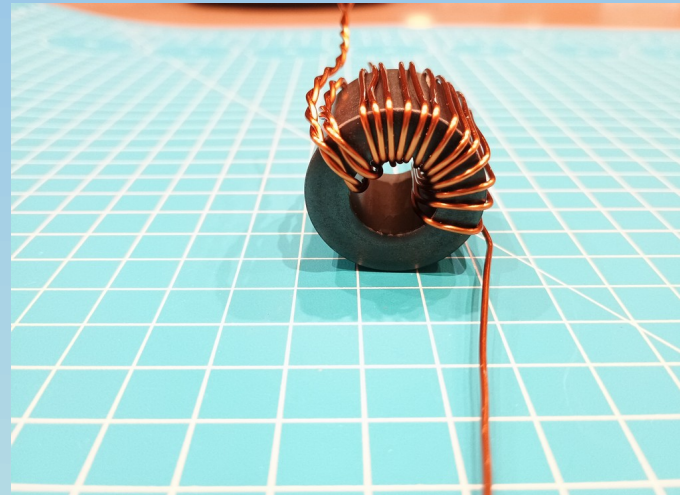
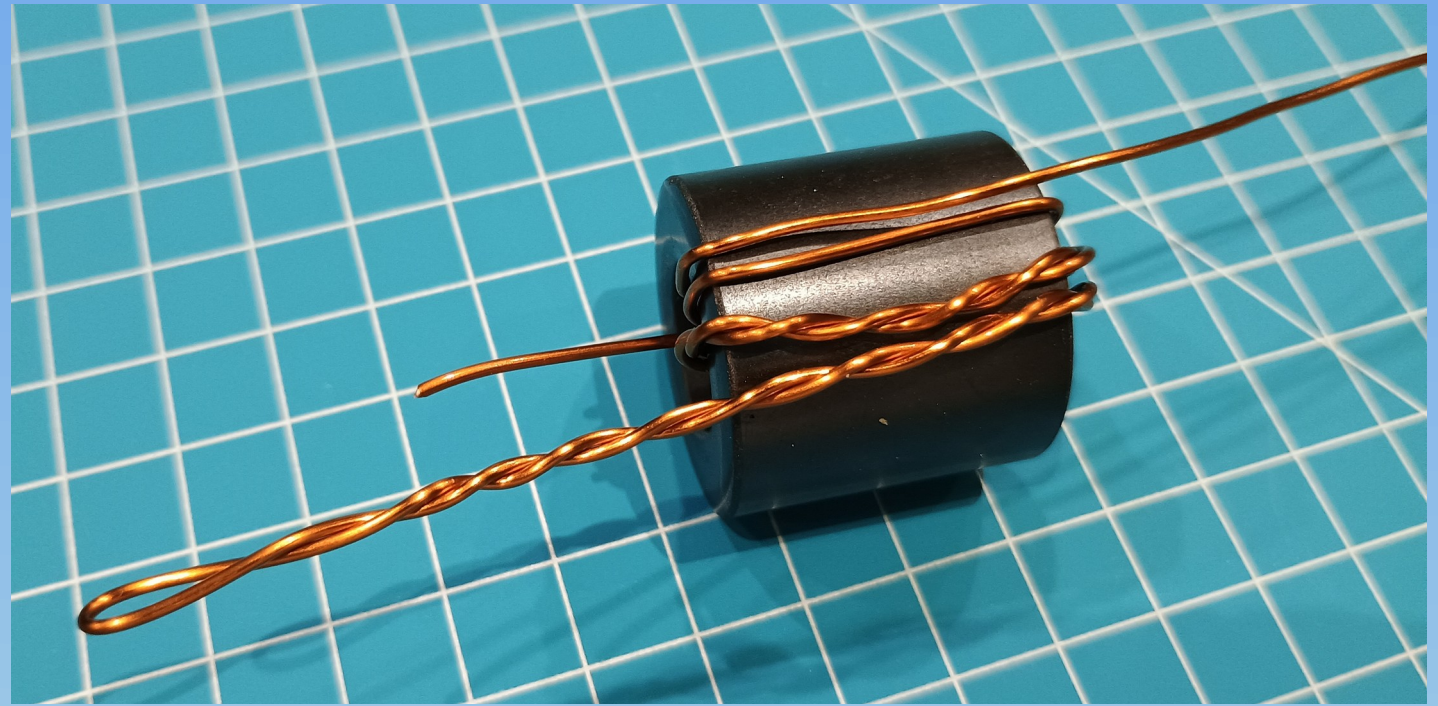
1) Built the Trafo

- Put the twisted wire trough the toroid and bend it down
- Now we havetwo windings for the primary (and secondary) side of the trafo



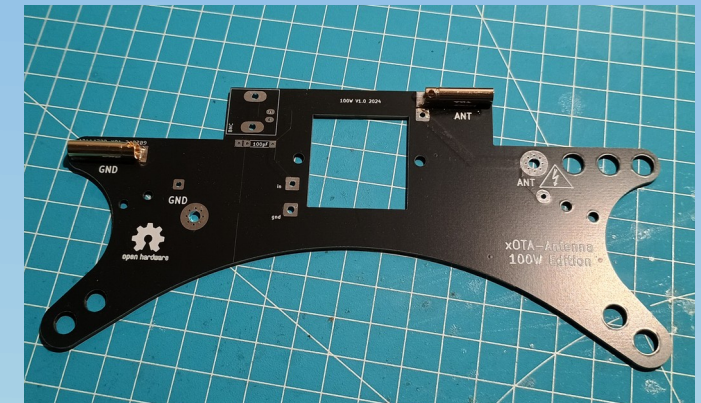
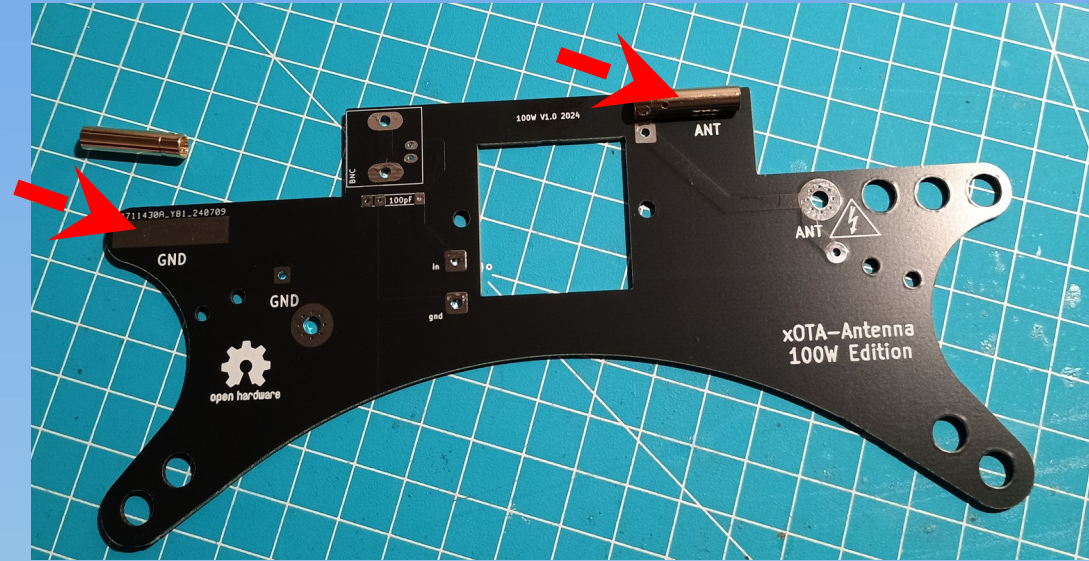
1) Built the Trafo

- Now, do 13 more windings with the long end of the wire to the left side



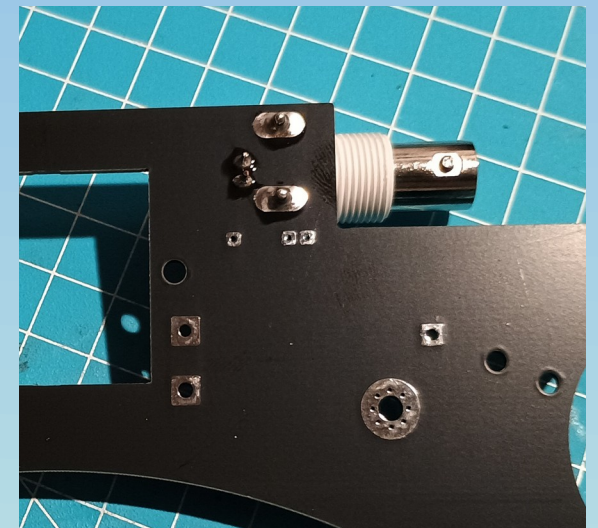
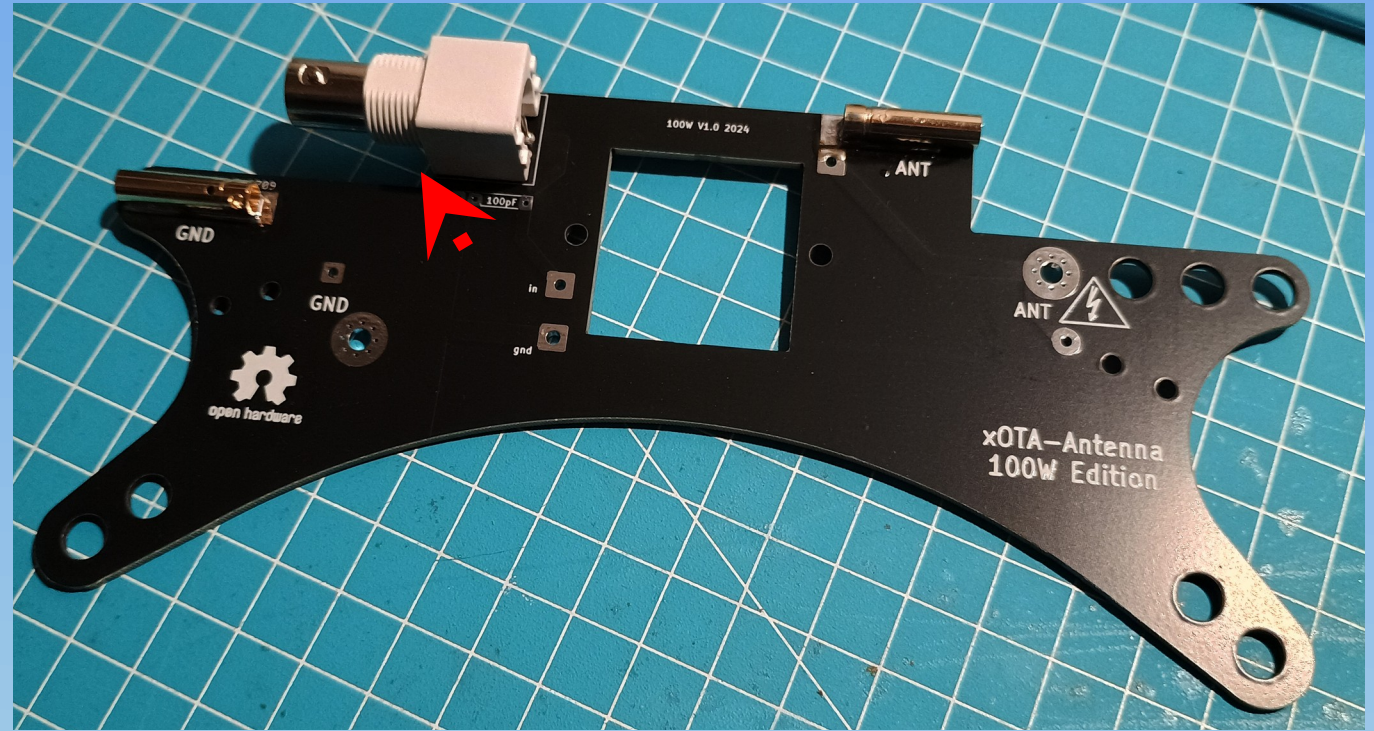
2) 4mm Sockets

If you want to use 4mm sockets for antenna wire connection (and counter poise, if so), solder them as first component to the PCB



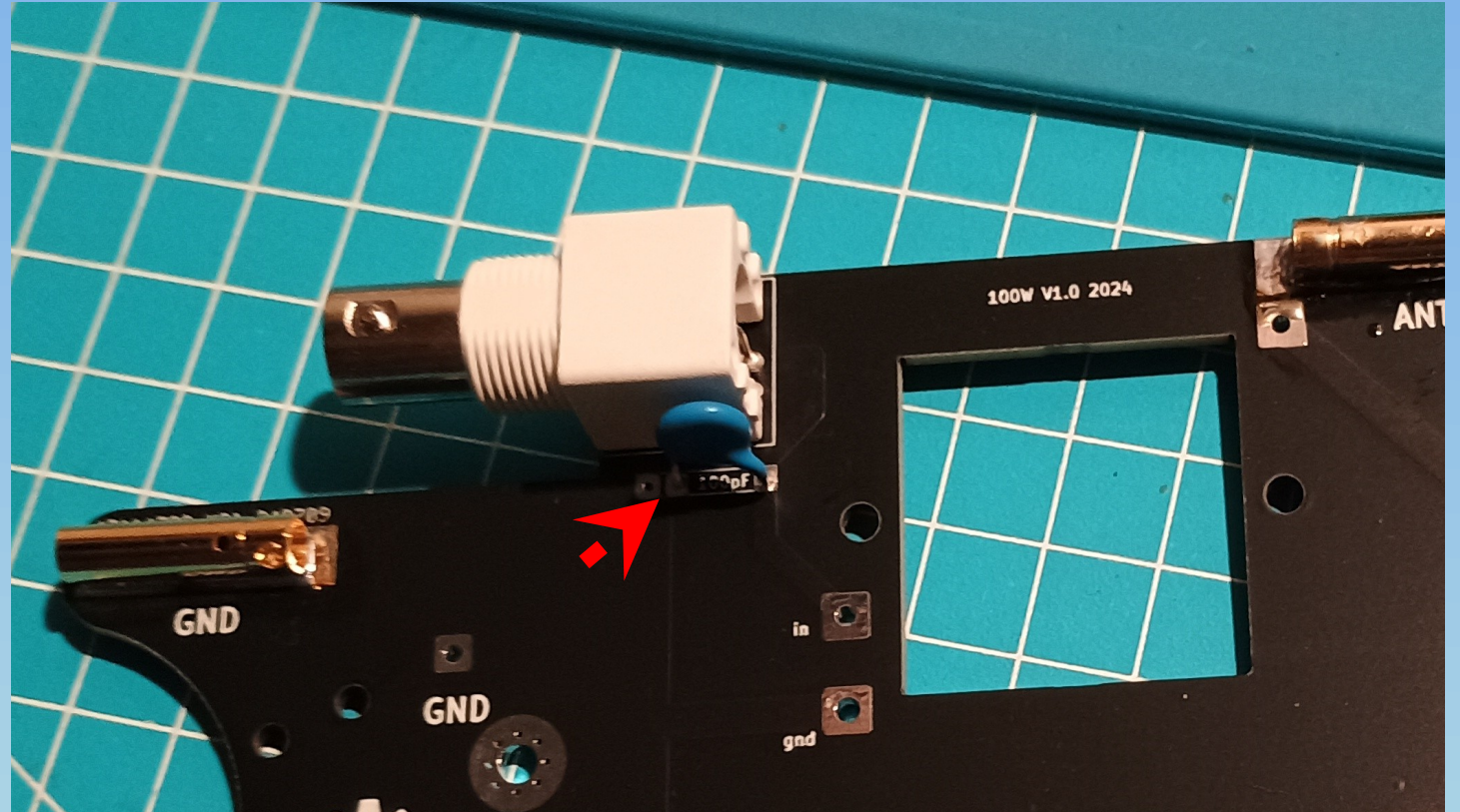
3) BNC Socket

- Solder BNC Socket to PCB



4) Capaciator

- solder 100pF ceramic capacitor to PCB



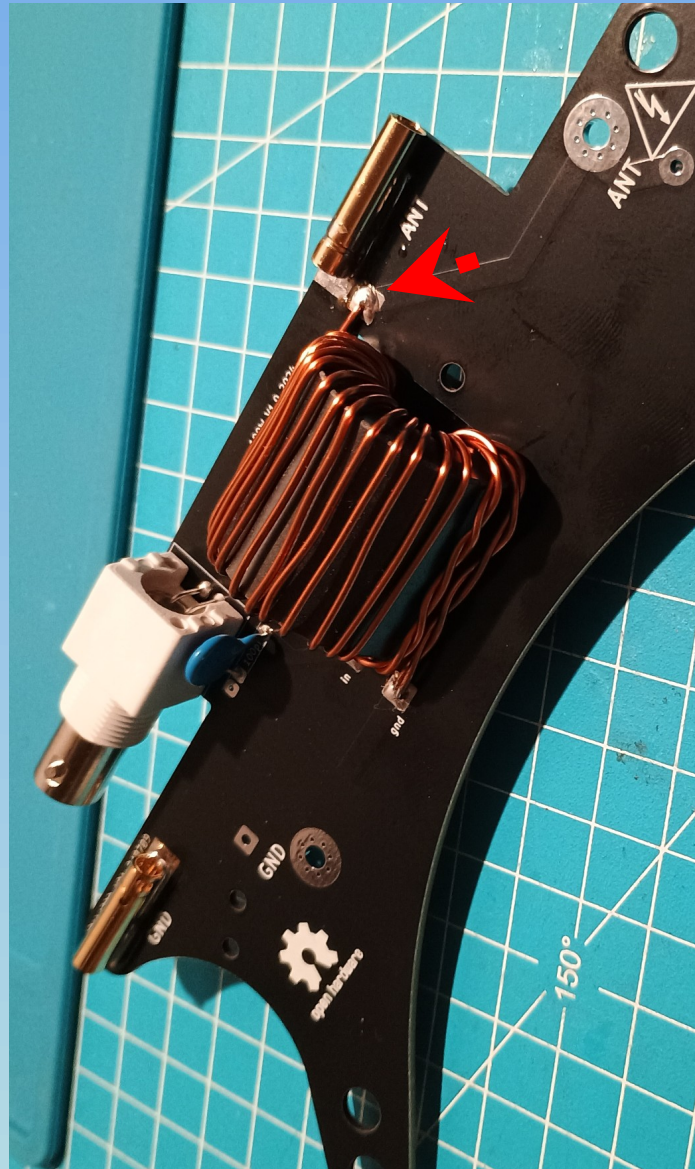
5) Insert Trafo

- Cut wire ends about 4cm
- Scrape laquer from the wire on the ends
- Do not tin the wire yet



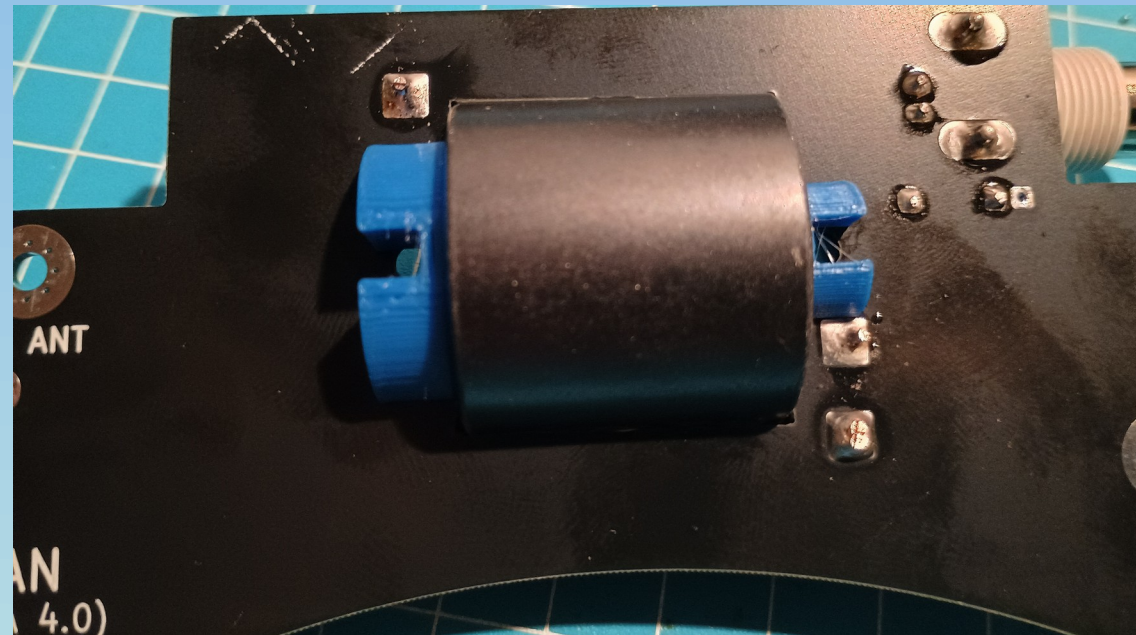
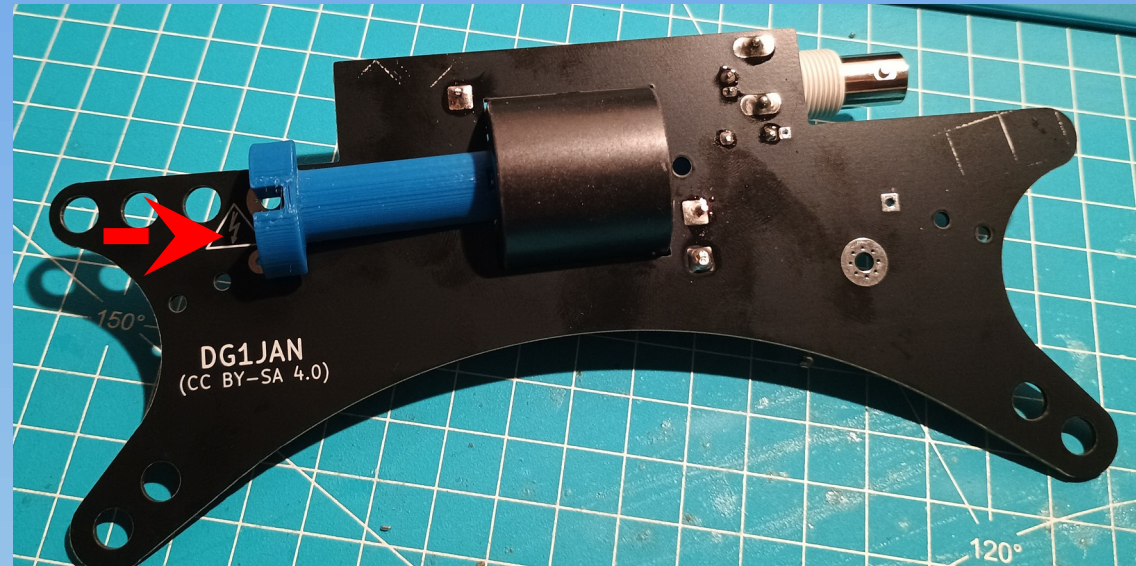
5) Insert Trafo

- Place trafo on the top side of the pcb
- Toroid must be in the middle of the pcb
- Solde wires according the pictures, there must be two wires in the GND Pad



6) insert wedge

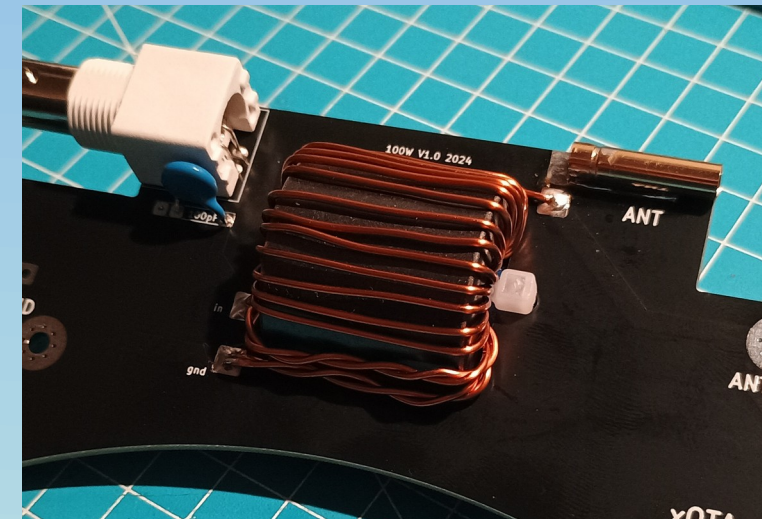
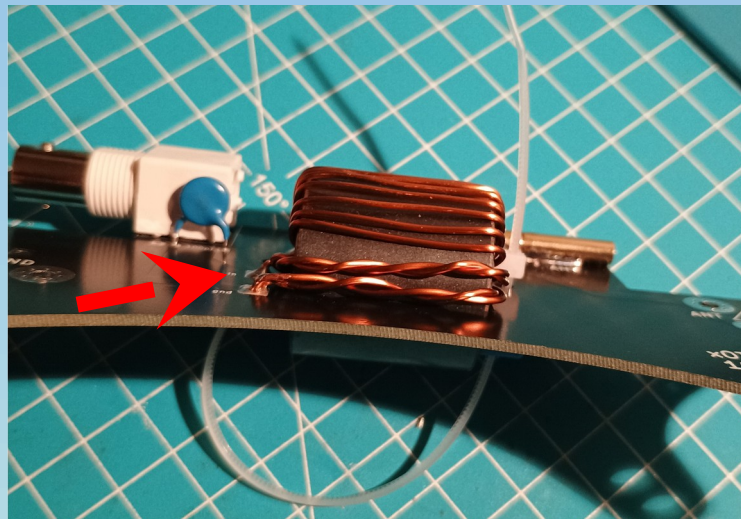
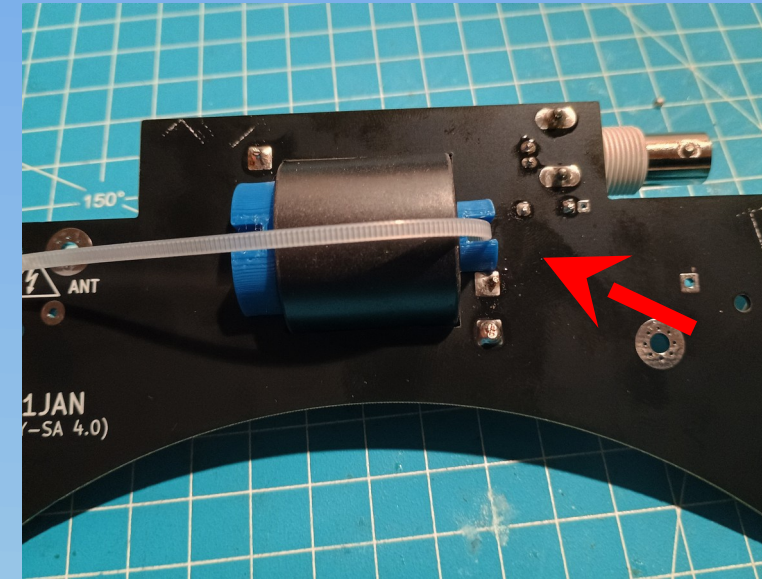
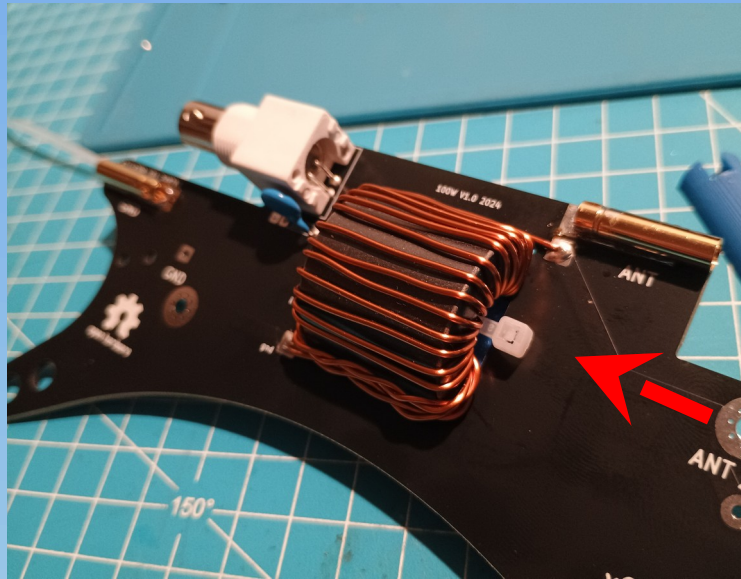
- Insert wedge on the bottom side of the pcb



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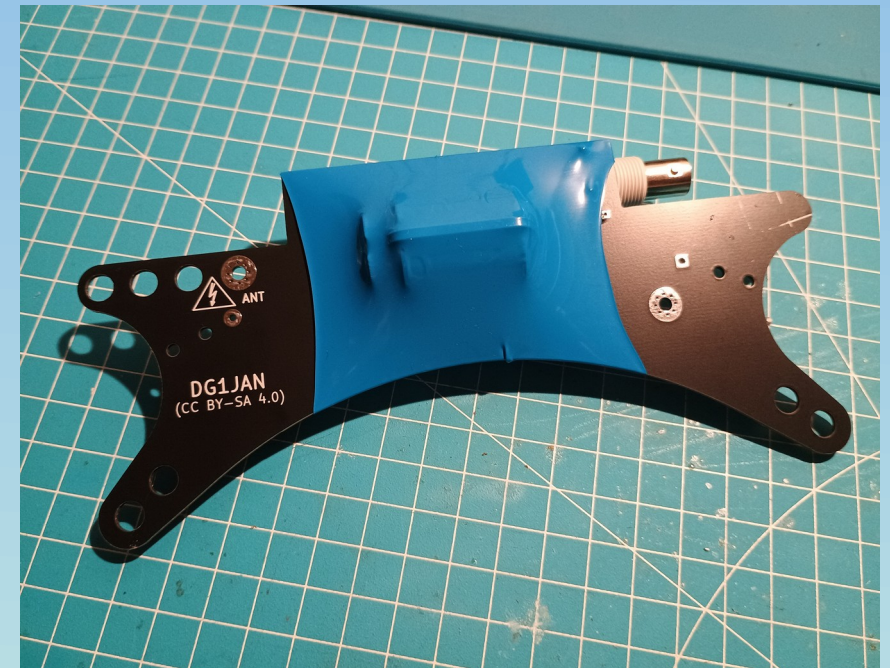
7) fixate trafo

- Use cable tie to secure trafo as shown in pictures:



8) heat shrink

- To have a basic protection of the trafo its recommend to use heatshrink



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